

1.Introduction(Structure and mechanics):

2.Theori:

2.1 Materials:

- **POM**
- **PEEK (PEEK, PEEK CF, PEEK GF)**
- **PLA**
- **PPS, PA-6 CF, PAHT-CF, PPA-CF**

2.2 Stresses

2.3 Tolerances

3.Method:

3.1 Solidworks:

There are multiple 3D modelling software programs available, but the one mainly used for this project was Solidworks. Like all similar programs, Solidworks is a software called CAD (Computer Aided Design) for drawing models in a 3D format. Its use of parametric design makes it effective and easy to use. Parametric design enables the designer/engineer to spot how adjustments to a drawing/model affect the assembly that's part of the drawing/model as a whole. In other words, it makes troubleshooting assemblies easy. Using this program, parts can be transferred to a 3D printer and printed out in a selection of materials, but most commonly some sort of plastic filament like PLA or ASA.

3.2 Vibration test:

Vibration testing is a critical part of the environmental qualification process for CubeSats, ensuring that the spacecraft can withstand the dynamic loads experienced during launch. During ascent, the launch vehicle generates significant vibrations caused by engine operation, aerodynamic forces, stage separation, and structural resonances. These loads can induce high stresses in the CubeSat structure, potentially leading to mechanical failure, loose fasteners, damaged electronics, or degraded payload performance.

To verify structural integrity, the CubeSat is mounted on an electrodynamic shaker table that reproduces the vibration environment expected during launch. Typical qualification tests include sine vibration tests, which identify structural resonances, and random vibration tests, which simulate the broadband vibration spectrum encountered during

launch. Throughout testing, accelerometers monitor the spacecraft's response to ensure that the structure, subsystems, and payload remain functional and undamaged.

Successful vibration testing demonstrates that the CubeSat satisfies launch qualification requirements and can safely endure the mechanical environment from liftoff to orbital deployment, reducing the risk of mission failure.

4.Results:

4.1 Choice of material:

Choosing the material for the structural parts and the battery holder for the cube satellite was important to make sure the parts would withstand the forces of a launch and be non-conducting, to avoid any electrical faults.

The structural parts included four side columns and two, top and bottom, main parts. To make the structure easier to produce, the design of the four side columns were identical. This also applied to the two, top and bottom, main parts.

During the concept period, all parts were 3D printed inhouse in PLA. This gave a fast and reliable way to test for sizes and tolerances in the real world. When the final design was approved, a material had to be chosen. For the structural parts, it was important that the material was strong, since they would be absorbing most of the forces at launch. With recommendations from external sources, PEEK would be a good candidate for the structural parts of the cube satellite. This is because PEEK can withstand high temperatures, has a high tensile strength, it is lightweight and highly electrically insulating.

Other variants of PEEK were also considered, like PEEK CF and PEEK GF. PEEK CF, with its carbon fiber filling, gave the highest strength but for the cost of being slightly electrically conductive and expensive. PEEK GF, with its glass fiber filling, was stronger than regular PEEK and maintained good electrical insulation, at a lower cost than PEEK CF. Ultimately, PEEK GF was chosen as the best alternative for the four side columns and two main parts.

The battery holder would not have to be able to manage the same load as the rest of the structure, and because of this, PLA was considered as a good choice at first. Ultem was also considered. These materials could be used in a 3D printer to make the production process cheap and fast, though a little more expensive with Ultem. External

sources recommended not using Ultem, but POM. Although POM had to be machined, it was stronger than PLA, and cheap. Other alternatives were also discussed, like PA-6 CF, PAHT-CF and PPA-CF. PA-6 CF shows a high tensile strength, seen in *Figure 4.1.1*, but since PA-6 CF is prone to brittleness and reduced layer adhesion, PAHT-CF and PPA-CF were given a closer look. Even though these two materials were recommended by retailers, the final choice of material was POM. This was because the material was both strong, cheap and offered good chemical resistance.

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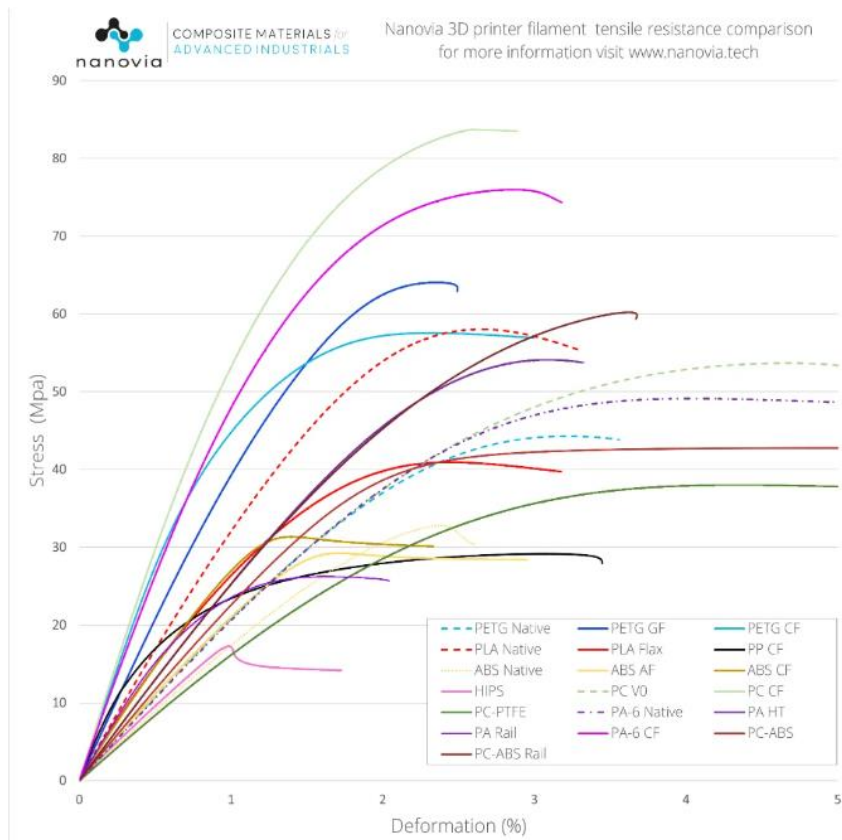


Figure 4.1.1: Plastic materials showing the relationship of stress and deformation.

4.2 Vibration test:

When the structure of the cube satellite was complete, it had to be fully assembled and tested. The satellite would have to sustain the forces that the rocket exerts on it, and to be prepared of the potential weak points and/or faults, a vibration test is crucial. To do this, the satellite was brought to Wallops Flight Facility in the United States, where they have a machine that can simulate the forces of a rocket in motion.

There were several stages of the vibration test, including:

Test as a standalone Cubesat

Tested integrated in the launcher.

Sine vibration testing at 10 g peak.
Random vibration testing at 12.7 grms.

Ultimately, the testing was successful and confirmed mechanical compatibility with the launch environment.

4.3 Assembly/Disassembly of cube satellite:

The satellite and its parts had to be assembled and disassembled several times to ensure that everything fits and works as intended. Doing this, often leads to discoveries that require either adjustments or new solutions.

The optimal placement for the batteries was in the middle of the satellite, between the internal pcb's. This was due to weight distribution, and space. To hold the batteries in place, a battery holder was made. The holder went through different iterations made with PLA, and with the final iteration, the part was made with POM. Because of the lower cost compared to other plastics, with the added benefit of strength and low electric conductivity, POM was the chosen material for the holder.

During the assembly of the battery holder with the batteries, an issue became apparent. When tightening the battery holder to the batteries, only two of the three batteries were locked in place. This was likely due to a fault in the design. Tightening the ends of the holder with the batteries would bend the middle part upwards, making the battery in the middle be looser than the other ones. To fix this, the batteries were taped to increase the thickness, with the middle one thicker than the other two. This made sure that all batteries stayed in place.

Making some final adjustments to the internal PCB, a new and last iteration was made. With the already assembled cubesat, it needed to be disassembled for the new PCB to replace the old one. In the process of replacing the PCB, it became apparent that distances between the connectors on the PCB did not correlate with the placement of the batteries and other pcbs. This made it impossible to install the new pcb. Taking some measurements, a new PCB was created. This PCB would fit perfectly in the assembly and be the final iteration.

4.4 3D modelling & technical drawing:

The structure, or frame of the satellite, began as sketches based on loose ideas. Once a decision was made on the design, it was used as a baseline to make a 3d model. Over time, the 3d model of the structure went through several changes, and the last iteration was modelled in Solidworks.

5.Discussions:

5.1.1 Why is PEEK GF better than PEEK?

When choosing a material for the structural parts of the cube satellite it was important that it could manage the forces exerted at launch. In this case PEEK is a good choice, since it has high mechanical strength, high thermal stability and resists beta, gamma, and x-rays. As a bonus it's also chemical resistant.

Even though PEEK has a lot of

6.Conclusion:

7.Abstract:

8.References:

Notater:

Solidworks, Teknisk tegning, 3d modellering, valg av materiale til struktur (PEEK-struktur, POM-battericase), toleranser, vekt basert på materiale (test på solidworks), sende inn tegninger av struktur til produksjon, demontere/bygge satellitt, jobbe med å feste batterier. Vibrasjonstest.

Peek: Fremsto som det beste alternativet.

Sidepaneler: FR4 – PCB kort.

*3d modellering, materialer, testing, software, tegninger, montering, konstruksjon.

Abstrakt, vibrasjonstest.